

NETWORKS AND COMPLEXITY

Solution 13-6

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 13.6: Abstract linear ODE system [Lvl. 2]

Consider the dynamical system

$$\begin{aligned}\dot{x} &= 2x - 9y \\ \dot{y} &= -x + 2y\end{aligned}$$

where the initial state is $x = y = 1$.

- a) Write the system in matrix form.

Solution

We write the system in the form

$$\dot{\mathbf{x}} = \begin{pmatrix} 2 & -9 \\ -1 & 2 \end{pmatrix} \mathbf{x} \quad (1)$$

where $\mathbf{x} = (x, y)^T$.

- b) Write the initial state as a linear combination of the eigenvectors of the matrix that appears in (a).

Solution

We write the characteristic polynomial

$$0 = (2 - \lambda)^2 - 9 \quad (2)$$

$$\lambda = 2 \pm 3 \quad (3)$$

We find the first eigenvector from

$$\begin{pmatrix} 2 & -9 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ a \end{pmatrix} = 5 \begin{pmatrix} 1 \\ a \end{pmatrix} \quad (4)$$

the first line reads

$$2 - 9a = 5 \quad (5)$$

which means $a = -1/3$, so the eigenvector for $\lambda_1 = 5$ is

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1/3 \end{pmatrix} \quad (6)$$

Because an eigenvector stays an eigenvector if we rescale it, we can multiply it by 3 which gives a nicer version of the same eigenvector

$$\begin{pmatrix} 3 \\ -1 \end{pmatrix} \quad (7)$$

but for the purpose of this exercise we will stick with the $\mathbf{v}_1$ that we found above.

For the second eigenvector we write

$$\begin{pmatrix} 2 & -9 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ a \end{pmatrix} = -1 \begin{pmatrix} 1 \\ a \end{pmatrix} \quad (8)$$

the first line reads

$$2 - 9a = -1 \quad (9)$$

and hence $a = 1/3$. So the eigenvector for $\lambda_2 = -1$ is

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1/3 \end{pmatrix} \quad (10)$$

We now need to write the initial condition as a combination of eigenvectors. We are looking for c_1 and c_2 such that

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -1/3 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1/3 \end{pmatrix} \quad (11)$$

- c) Solve the system (i.e. compute $x(t)$ and $y(t)$).

Solution

From the first row in (11) we can see that

$$c_1 + c_2 = 1. \quad (12)$$

and hence

$$c_2 = 1 - c_1 \quad (13)$$

From the second row we get

$$\frac{1}{3}c_2 - \frac{1}{3}c_1 = 1 \quad (14)$$

We multiply the equation by 3 which gives

$$c_2 - c_1 = 3 \quad (15)$$

Substituting $c_2 = 1 - c_1$ from above yields

$$1 - 2c_1 = 3 \quad (16)$$

Solving this we find

$$c_1 = -1 \quad (17)$$

And using either of the relations above

$$c_2 = 2 \quad (18)$$

We know from the lecture that the solution has the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2. \quad (19)$$

So, in this case

$$\mathbf{x}(t) = -e^{5t} \begin{pmatrix} 1 \\ -1/3 \end{pmatrix} + 2e^{-t} \begin{pmatrix} 1 \\ 1/3 \end{pmatrix}. \quad (20)$$